Tobias Mwangi-Hartley

[EMAIL] | [PHONE] | [LINKEDIN]
Data Scientist | Applied Machine Learning

PROFESSIONAL SUMMARY

Data Scientist with experience applying machine learning and statistical modelling to demand forecasting, churn propensity and experimentation. Built Python workflows for scheduled forecasting and weekly batch scoring, and a Streamlit experiment-planning tool used by product and growth teams. Seeking to deepen production ML engineering practice while contributing applied modelling, data analysis and testing skills.

TECHNICAL SKILLS

Programming and libraries: Python, SQL, R, pandas, scikit-learn, LightGBM, XGBoost, statsmodels, Streamlit
Machine learning and statistics: Demand forecasting, churn propensity modelling, feature engineering, model evaluation, SHAP, Bayesian hierarchical models, experimental design, sequential A/B testing
Data and platforms: BigQuery, dbt, Vertex AI Workbench, Looker, R Markdown, Docker, pytest
Additional learning: PyTorch and torchvision (course project)

EXPERIENCE

Data Scientist | Thistlebox | 2022–Present

- Developed a weekly demand forecast for orders by recipe and region, six weeks ahead, using LightGBM and calendar, price and menu-position features.
- Improved forecast WAPE from 23% to 14%; estimated food waste reduction of 9%. The forecast runs as a scheduled Vertex AI Workbench notebook, writes to BigQuery, and is used by operations in Looker.
- Built an XGBoost churn propensity model scored weekly in BigQuery; retention offers are sent to top-decile customers. In a 10% holdout validation, churn was down 2.1 percentage points in the treated group.
- Built a Streamlit app for planning and analysing sequential A/B tests, used by approximately 25 people in product and growth. Wrote its Dockerfile; the data platform team hosts it on Cloud Run and handles deployment.
- Since 2025, written pytest unit tests for forecasting feature code, reaching approximately 60% coverage.

Junior Data Analyst | Firth Energy Retail | 2020–2022

- Segmented customers and analysed tariffs using R and SQL.
- Automated monthly regulator reports with R Markdown, saving the team approximately two days each month.

EDUCATION

MSc in Statistics
Dissertation: Bayesian hierarchical models

ADDITIONAL PROJECT

Image classification course project | 2025
Fine-tuned a small CNN in PyTorch on a public food images dataset, achieving approximately 80% accuracy. Learning project completed as part of an online deep learning course.